Diagnostic Report generated by gemini-2.5-pro

Diagnostic Analysis of Run_37

1. Summary of Key Deviations

The analysis of Run_37 reveals a physically contradictory set of symptoms, pointing towards a specific type of component failure rather than a simple blockage or leak.

- **Flow_Bypass (Moderate):** A moderate **negative offset** of **-10% to -14%** is observed, which is unusual.
- **Flow_Pump (Minor):** A corresponding minor **negative offset** of around **-9%** is seen on the pump flow.
- **Flow_Engine1-4 (Minor):** All engine flows show a consistent **negative offset** of **-3% to -5%**.
- **p_Combustion (Minor):** In direct contradiction to the loss of flow, the combustion pressure shows a consistent **positive offset** of approximately **+8%**.
- **Other Signals (Nominal):** All other pressure and flow signals show only negligible deviations.

2. Root-Cause Hypotheses

Rank	Suspected Component	Fault Type	Supporting Evidence
1	**Bypass Valve**	**Stuck Partially Closed / High Cracking Pressure**	This is the most compelling hypothesis. A bypass valve that is stuck partially closed or has a higher-than-normal cracking pressure would fail to open correctly. This would force more fuel than intended downstream towards the engine, causing the back-pressure (`p_Combustion`) to rise. This increased resistance would, in turn, cause the overall pump flow (`Flow_Pump`) to drop, and the flow through the bypass line (`Flow_Bypass`) would also be lower than normal. This single fault elegantly explains the contradictory symptoms of low flow and high pressure.

2	**Main Fuel Pump & p_Combustion Sensor**	**Dual Fault: Pump Under-performance & Sensor Error**	It's possible the pump is failing (explaining the low flows) and, coincidentally, the `p_Combustion` sensor has a positive offset error (explaining the high pressure). While possible, a single physical fault (Hypothesis 1) that explains all symptoms is more likely.
3	**Fuel Metering Unit (FMU)**	**Partial Blockage**	A partial blockage at the FMU would explain the high `p_Combustion` and low `Flow_Engine` readings. However, it would typically cause `Flow_Bypass` to *increase* as flow is diverted, whereas here the bypass flow is *decreasing*. This makes a simple blockage unlikely.
4	**Main Fuel Pump**	**Moderate Degradation**	Pump degradation would explain the drop in `Flow_Pump` and `Flow_Bypass`. However, it cannot explain why the downstream `p_Combustion` would *increase*. A failing pump should lead to lower, not higher, downstream pressure.
5	**Multiple Sensor Failures**	**Various Errors**	It is highly improbable that `Flow_Pump` and `Flow_Bypass` sensors would fail with a negative offset while the `p_Combustion` sensor simultaneously fails with a positive offset. The pattern is more indicative of a single, counter-intuitive physical fault.

3. Most Probable Hypothesis

The most probable root cause is that the **Bypass Valve is Stuck Partially Closed or has a high cracking pressure.**

Reasoning: This single mechanical failure is the most parsimonious explanation for the otherwise contradictory data. The core of the issue is explaining how flows can be low while pressure is high. A bypass valve that fails to open properly does exactly this. It essentially creates a partial blockage in the bypass return loop, forcing more fluid into the downstream part of the system than the controls intend. This unwanted downstream flow increases the resistance and back-pressure (`p_Combustion` is high). In response to this higher-than-expected system resistance, the fixed-displacement pump's total output flow drops (`Flow_Pump` is low), and because the valve isn't opening, the measured `Flow_Bypass` is also low. This single event explains the entire symptom set.